

Valencia Flood Mitigation Plan 2024

Mitigation Strategies and Action Plan

Generalitat Valenciana - Conselleria d'Emergencia Climatica

1. Strategic Priorities Post-2024 DANA

Following the catastrophic October 2024 DANA event, the Generalitat Valenciana has established five strategic priorities for flood risk reduction:

Priority 1 - Structural Flood Defense: Construction of retention basins, channel improvements, and flood walls in the most affected areas (barranco del Poyo, Magro river, southern L'Horta).

Priority 2 - Early Warning Enhancement: Upgrading the SAIH and AEMET systems with higher-density radar coverage, improved DANA trajectory forecasting, and cell-broadcast emergency alerts.

Priority 3 - Land Use Planning Reform: Stricter enforcement of PATRICOVA zones, prohibiting new residential development in Zone A1, and requiring flood-resilient construction in Zone A2.

Priority 4 - Nature-Based Solutions: Restoring natural flood retention capacity through reforestation of headwater catchments, wetland restoration, and sustainable urban drainage.

Priority 5 - Community Resilience: Public awareness campaigns, community emergency plans, flood insurance reform, and post-disaster recovery protocols.

2. Planned Infrastructure Projects

The following major infrastructure projects are planned or underway (2024-2030):

Project: Barranco del Poyo Flood Defense (Budget: 850M EUR)

Description: Construction of upstream retention basins, channel widening through populated areas, and elevated flood walls. Designed to handle 1-in-500-year flow rates.

Comarcas affected: L'Horta Sud, Valencia

Timeline: 2025-2029

Project: Xuquer River Basin Retention (Budget: 420M EUR)

Description: Three new flood retention reservoirs upstream of Alzira and Algemesi to reduce peak flows in La Ribera Baixa during extreme rainfall events.

Comarcas affected: La Ribera Alta, La Ribera Baixa

Timeline: 2025-2028

Project: Segura River Capacity Upgrade (Budget: 280M EUR)

Description: Increasing channel capacity from 400 to 800 m³/s through Orihuela and the Vega Baja, with new overflow detention areas.

Comarcas affected: La Vega Baja

Timeline: 2025-2027

Project: Coastal DANA Drainage Network (Budget: 190M EUR)

Description: Upgraded storm drainage in La Marina Alta and La Safor coastal towns, including underground retention tanks and improved outfall capacity.

Comarcas affected: La Marina Alta, La Safor, La Marina Baixa

3. Early Warning and Emergency Response

Enhanced early warning system components:

AEMET DANA Forecasting: Upgraded numerical weather prediction models with 1km resolution for the western Mediterranean, providing 48-72 hour advance warning of DANA formation.

SAIH Xuquer Enhancement: 50 additional automated rainfall and flow gauging stations, particularly in the barranco del Poyo, Magro, and Albaida subcatchments.

ES-Alert Cell Broadcast: Implementation of the EU-mandated cell broadcast emergency alert system, capable of reaching all mobile phones in affected zones within 60 seconds. Operational since December 2024.

Municipal Emergency Plans: All 542 municipalities in the Comunitat Valenciana required to update flood emergency plans by December 2025, including designated evacuation routes, shelter locations, and vulnerable population registries.

Automated Flood Barriers: Installation of automatic rising flood barriers at 45 critical underpasses and low-lying road crossings in high-risk comarcas (2025-2026).

4. EU Floods Directive Compliance

Under EU Directive 2007/60/EC (Floods Directive), Spain must:

- Complete Preliminary Flood Risk Assessment (PFRA) every 6 years (next: 2025)
- Produce Flood Hazard and Risk Maps for all significant flood risk areas
- Develop Flood Risk Management Plans (PGRI)
- Report progress to the European Commission

The Comunitat Valenciana's PGRI for the Xuquer Hydrographic Demarcation (2022-2027) identifies 89 Areas of Potentially Significant Flood Risk (ARPSIs). Following the 2024 DANA, an emergency revision has been initiated to add 12 new ARPSIs and upgrade risk classifications for 23 existing areas.

EU Recovery Funds: Spain has allocated 3.2 billion EUR from NextGenerationEU and EU Solidarity Fund specifically for Valencia flood reconstruction and resilience improvements (2025-2027).

5. Nature-Based Solutions

Green infrastructure and natural flood management measures:

Headwater Reforestation (Budget: 85M EUR): Planting 12,000 hectares of native forest in upper catchments of the Xuquer, Turia, and Magro rivers to increase rainfall interception and reduce surface runoff by 15-20%.

Wetland Restoration (Budget: 65M EUR): Restoring 3,500 hectares of coastal wetlands in L'Albufera, the Marjal de Pego-Oliva, and the Marjal dels Moros as natural flood buffers.

Agricultural Water Retention (Budget: 40M EUR): Working with farmers in La Ribera Alta and La Ribera Baixa to implement controlled flooding of rice paddies and citrus groves during extreme events, providing temporary flood storage.

Urban Green Infrastructure (Budget: 95M EUR): Rain gardens, bioswales, green roofs, and permeable paving in Valencia city, Gandia, Alzira, and Orihuela to reduce urban runoff volumes by 30% for events up to 50mm/hour.